

Srinivas Sivaratri

AI Engineer | LLM evaluation, model debugging, developer tools

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srinivassivaratri.is-a.dev | Remote, India (IST, UTC+5:30) | Open to full-time and contract

Summary

AI engineer with an MS in Computer Science. Builds LLM systems and the evaluations that check them, and publishes the numbers, including the ones that lose to a baseline. Upstream work reviewed by better-auth and go-ethereum maintainers.

Skills

AI and LLM: LLM evaluation, RAG, hybrid retrieval (BM25, Reciprocal Rank Fusion, cross-encoder reranking), token-level tracing and replay, LoRA fine-tuning, diffusion models

Machine Learning: PyTorch, Hugging Face Transformers, PEFT, sentence-transformers

Languages: Python, TypeScript, Go, JavaScript, SQL, C

Engineering: FastAPI, Flask, Next.js, React, Node.js, Bun, PostgreSQL, Redis, SQLite, Docker, GitHub Actions, pytest, Vitest

Experience

Software Engineer, Independent | Remote May 2025 to present

- Traced an email-enumeration leak in better-auth to its Have I Been Pwned plugin and proposed a request-scoped skip flag with a regression test. Maintainers adopted the approach and shipped it (PR #5998).
- Designed "geth db truncate-freezer" for go-ethereum (PR #31351), revised across three core-maintainer review rounds to batch-process instead of holding about 15 GB in memory. The core team later built its own version.

Software Engineer, Accenture | Bangalore, India July 2021 to July 2022

- Wrote a Python ETL script that automated data processing, saved 50 labor hours per month, and cut errors.
- Implemented a GitHub Actions CI/CD pipeline.

Projects

unmaze | Python, PyTorch, NumPy github.com/sivaratrisrinivas/unmaze

- Built a small diffusion model that solves hedge mazes by denoising random noise into the path from entrance to centre. A 2.7M-parameter U-Net with self-attention and a 50-step DDIM sampler trained in about 50 minutes on 3 CPU threads. A strict judge accepts only one exact simple path and runs no repair step.
- Solved 4,990 of 5,000 fresh 11x11 mazes it had never seen (99.8%), then audited the number. On depth-first-search mazes it never trained on it solves 66%, all 10 failures were in the longest quarter of routes, and the 300 longest routes of 30,000 score 93%.

Bingo | Python, NLTK, sentence-transformers, CLIP, Gemini github.com/sivaratrisrinivas/Bingo

- Built a command-line movie search over 5,000 films during Boot.dev's RAG course, with BM25 over an inverted index, chunked embeddings, Reciprocal Rank Fusion, cross-encoder reranking, image search, and LLM answers with citations.
- Built an 834-query known-item eval. On the test split, RRF with cross-encoder reranking put the target movie in the top 10 91.6% of the time against 86.3% for BM25, and plain RRF lost to BM25 alone. Rewrote BM25 to score from precomputed term statistics, cutting p95 latency from 3.4 s to 5 ms.

Education

MS, Computer Science, University of Alabama at Birmingham, AL, USA Aug 2023 to May 2025

BS, Computer Science, BV Raju Institute of Technology, India Aug 2017 to June 2021

Boot.dev, coursework, HTTP Servers in Go, RAG, CI/CD with GitHub Actions and Docker, Memory Management in C, SQL